Avocado Price Predictor: Technical Report

ID: 242031013

Abstract

A tuned Random Forest predicts the weekly average price of an avocado with a typical error (MAE) of about $0.21 on the newest 20% of weeks (R^2 = 0.467), and a simple Linear Regression baseline is almost as good (MAE $0.217, R^2 = 0.460).

The task is regression: given the week, region, type (conventional or organic) and sales volumes, estimate `AveragePrice`. The data are 18,249 weekly retail records from the Hass Avocado Board (Kaggle "Avocado Prices"), covering 54 regions from January 2015 to March 2018. The data were split by time, so the model is always tested on weeks that come after the training weeks, which avoids leakage from the future. Four models were compared: a Dummy mean predictor, Linear Regression, a default Random Forest and a Random Forest tuned with RandomizedSearchCV and time-ordered cross-validation.

All three real models clearly beat the Dummy predictor (test RMSE 0.423), but they are close to each other (test RMSE 0.279 to 0.290). All models underestimate the unusually high prices of autumn 2017. The model estimates a price from same-week market data; it is not a forecast of future prices.

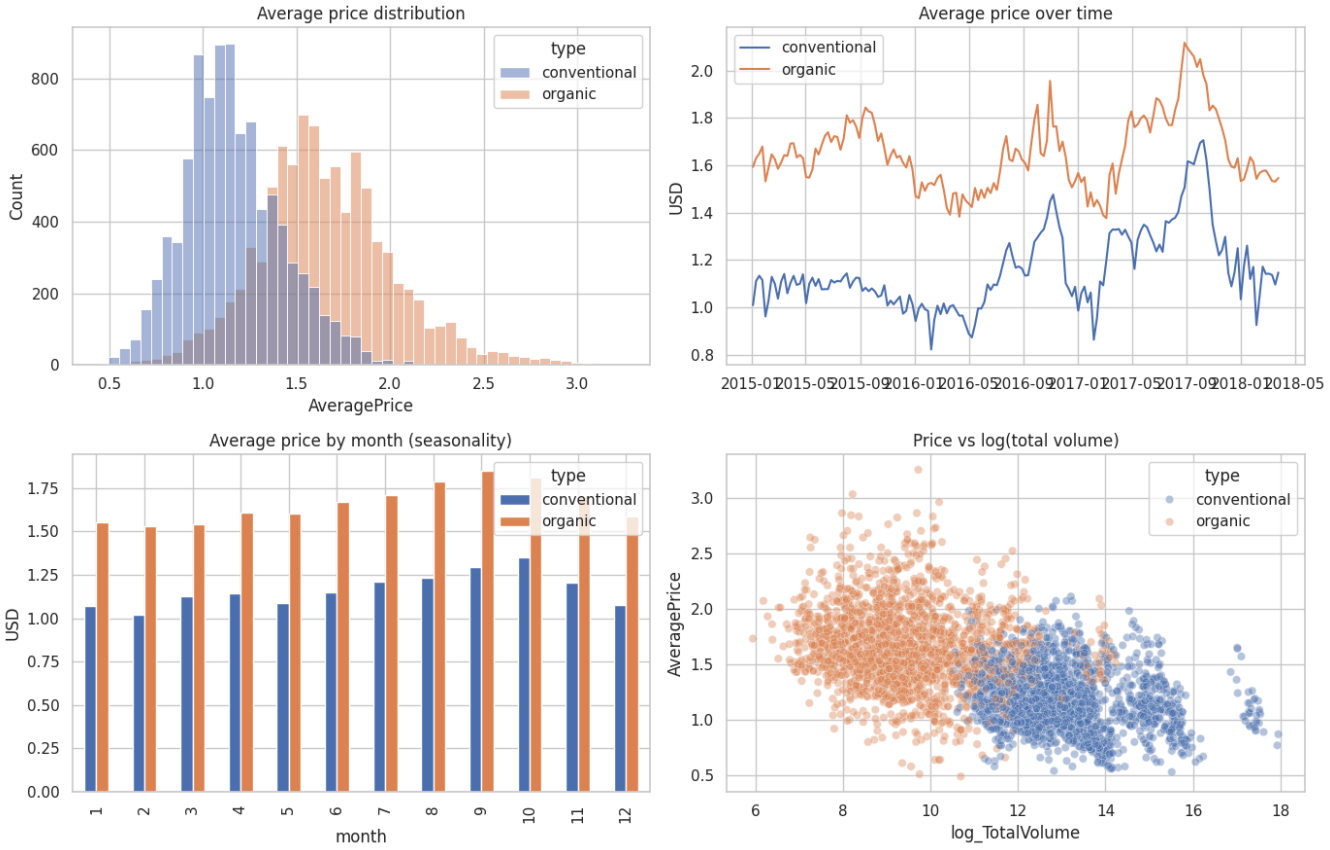
Data Exploration

The dataset is clean, and the main drivers of price are the avocado type, the time period and the sales volume.

Item	Value
Source	Hass Avocado Board weekly retail data (Kaggle "Avocado Prices")
Rows	18,249
Period	2015-01-04 to 2018-03-25 (weekly)
Regions	54 (some are larger areas that contain other regions, such as TotalUS and West)
Types	conventional, organic
Target	`AveragePrice` (USD per avocado): mean 1.41, min 0.44, max 3.25

Item	Value
Missing values / duplicate rows	none / none removed

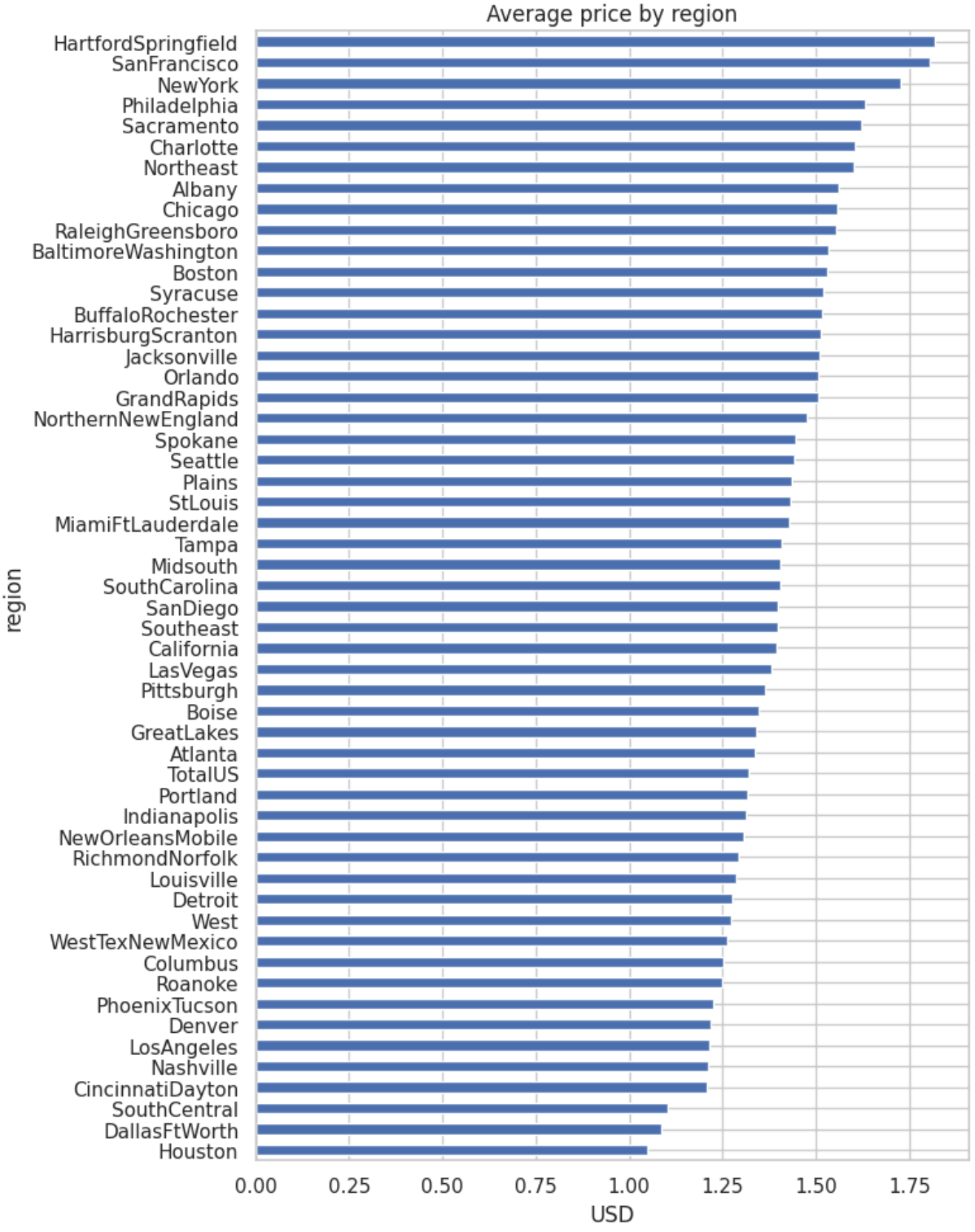
Sales volumes are very skewed: the mean of `TotalVolume` is about 850,644, the median about 107,377 and the maximum about 62.5 million. For this reason all volume columns were converted with log(1 + x).



What the overview charts show:

1. **Organic avocados cost more.** Conventional prices are centred at about $1.1 and organic prices at about $1.6 to $1.8. The correlation between `is_organic` and price is 0.616.
2. **The price level changes over time.** Conventional prices stay near $1.0 to $1.1 in 2015 and rise to about $1.7 in autumn 2017; organic prices peak at about $2.1 in the same period. The newest weeks therefore have a higher mean price ($1.529) than the training weeks ($1.375).
3. **There is a seasonal pattern, but it must be read carefully.** Monthly averages are highest in September and October and lowest in the first months of the year. The averages mix several years and the large 2017 spike lies in autumn, so part of the pattern is probably that one spike. The data end in March 2018, so months 1 to 3 contain four years and the other months three.

4. **Higher volume goes with lower price.** `log_TotalVolume` has the strongest negative correlation with price (-0.583). Part of this is the type effect: organic avocados sell in small volumes at higher prices. Correlation does not prove that volume causes the price.



5. **Prices differ by region.** HartfordSpringfield, SanFrancisco and NewYork have the highest average prices (about $1.7 to $1.8); Houston, DallasFtWorth and SouthCentral the lowest (about $1.05 to $1.1). The values in this paragraph are read from the chart and are approximate.

Methodology

The data are split by time, all preprocessing is learned from the training weeks only, and four models are compared on the same test weeks.

Preprocessing and features

- Dates were converted to dates; rows with missing values, duplicate rows and rows with price of 0 or less were removed (no row was removed: 18,249 of 18,249 kept).
- From the date: `year`, `month` and `weekofyear`. The type became `is_organic` (0 or 1).
- The eight volume columns were transformed with log(1 + x), because they are very skewed.
- 13 features in total: 4 calendar/type features, 8 log-volume features and `region` (one-hot encoded, 54 categories).

Time-based split (to avoid data leakage)

Set	Rows	Period	Mean price
Train (oldest 80% of the dates)	14,577	2015-01-04 to 2017-07-30	$1.375
Test (newest 20% of the dates)	3,672	2017-08-06 to 2018-03-25	$1.529

A random split would let the model see weeks from the future, which would give overly optimistic results. The test set was not used for training or tuning.

Models

Model	Role	Details
Dummy	Minimum to beat	Always predicts the mean training price
Linear Regression	Simple baseline	Numeric features scaled with StandardScaler, region one-hot encoded, inside a Pipeline
Random Forest (default)	Complex model	150 trees, other settings default

Model	Role	Details
Random Forest (tuned)	Final model	RandomizedSearchCV, 12 trials

Hyperparameter tuning

- RandomizedSearchCV with 12 random trials and `TimeSeriesSplit` with 3 folds (each fold trains on earlier weeks and validates on later weeks); score = negative RMSE.
- Search space: `n_estimators` 100 / 200 / 300, `max_depth` 8 / 12 / 16 / none, `min_samples_leaf` 1 / 2 / 5 / 10, `max_features` sqrt / 0.5 / 1.0.
- Best setting: 100 trees, `max_depth` 12, `min_samples_leaf` 2, `max_features` 0.5. Best cross-validated RMSE: 0.265.

Metrics

- RMSE: typical error in USD, large errors count more.
- MAE: average absolute error in USD.
- R^2: share of the price variation explained; 0 means no better than predicting the mean price of the test weeks, and a negative value is worse than that (the Dummy model scores -0.152 because the test weeks have a higher mean price than the training weeks).
- Train RMSE is compared with test RMSE to detect overfitting.

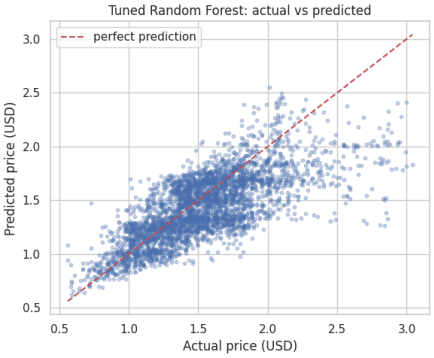
Results

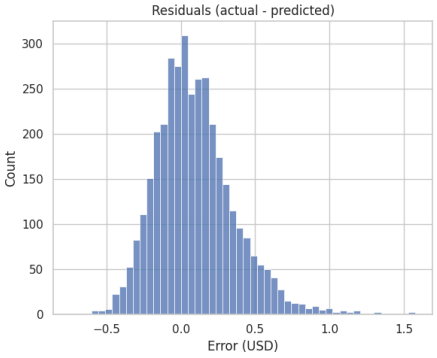
All three real models cut the Dummy test RMSE by about 31% to 34%, but they differ from each other by only about 0.01 USD.

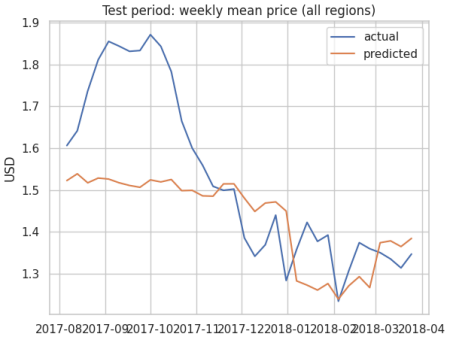
Model	Train RMSE	Test RMSE	Test MAE	Test R^2
Dummy (mean price)	0.399	0.423	0.322	-0.152
Linear Regression (baseline)	0.195	0.290	0.217	0.460
Random Forest (default)	0.042	0.279	0.209	0.498
Random Forest (tuned, final model)	0.114	0.288	0.213	0.467

- **Final model.** The tuned Random Forest was chosen because it had the best cross-validated RMSE on the training data (0.265). The default Random Forest scored slightly better on the test set (0.279), but choosing it now would mean selecting a model by looking at the test set, so it is reported but not used as the final model.

- **Are the differences meaningful?** The gap between the three real models is about 0.01 USD in RMSE. With one test period, I cannot say it is meaningful. A Random Forest and a Linear Regression perform about the same on this problem.
- **Overfitting.** The test RMSE divided by the train RMSE is about 6.6 for the default Random Forest, 2.5 for the tuned one and 1.5 for Linear Regression. Tuning reduced overfitting but did not improve the test error.
- **Error in practice.** The tuned model is on average about $0.21 away from the true price (MAE), which is about 14% of the mean test price of $1.529. The typical error (RMSE) is about $0.29.

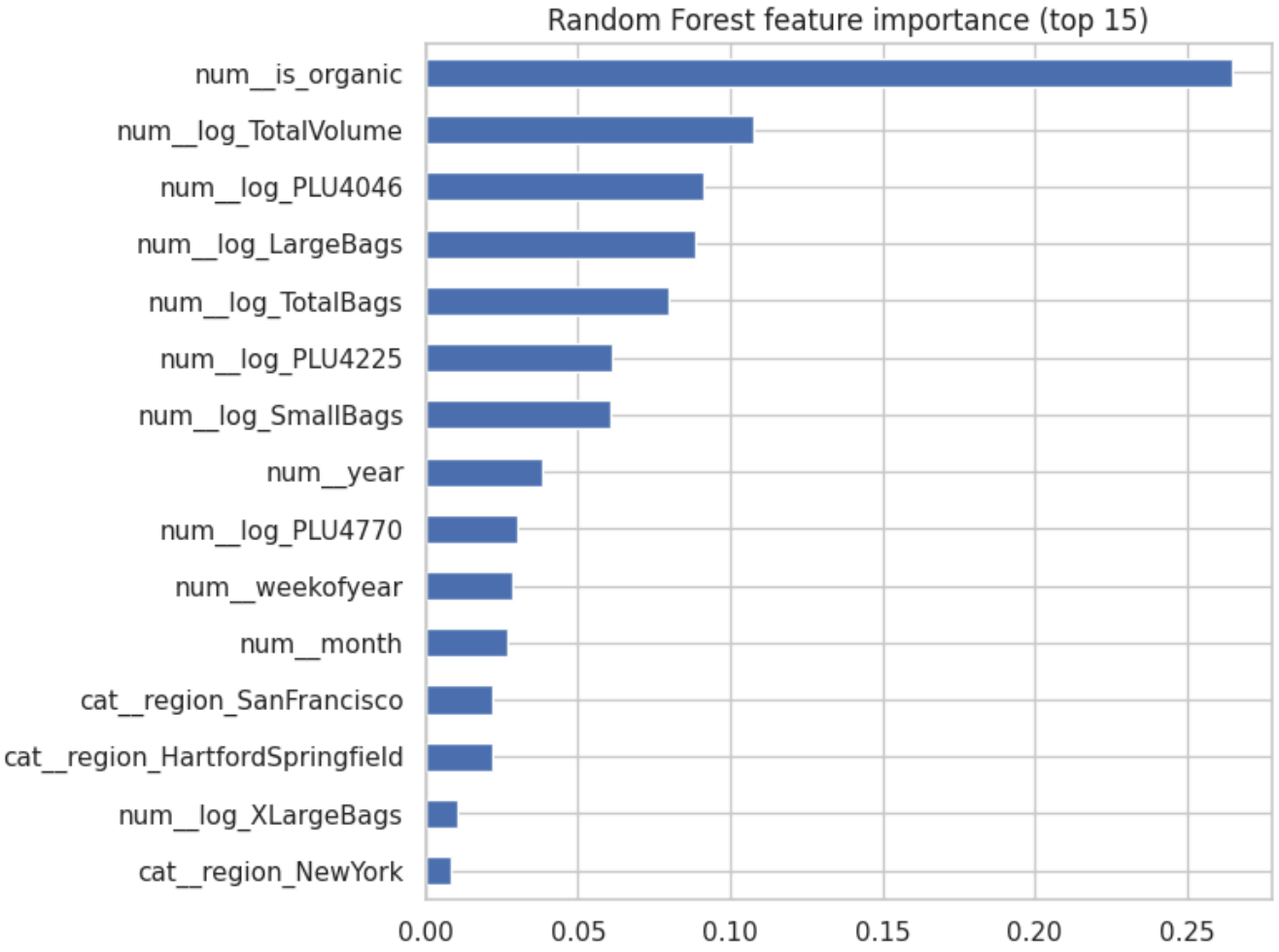






What the evaluation charts show

- Predictions follow the diagonal, but high prices (above about $2.5) are predicted too low. The residuals are centred near zero with a longer tail on the positive side (actual price higher than predicted).
- In the test period, the real weekly mean price was about $1.85 in August to October 2017 while the model predicted about $1.52. From December 2017 the prediction is much closer to the real price. The model does not capture the 2017 price spike, which is probably caused by factors that are not in the data.



Feature importance. `is_organic` is the most important feature (about 0.27), followed by the volume features (`log_TotalVolume` about 0.11). Region and calendar features are less important. These are impurity-based importances read from the chart, so the values are approximate, and correlated volume features share their importance.

Demo prediction. For New York, conventional avocados, week of 2018-03-25, the model predicted $1.60 and the actual price was $1.34 (an error of $0.26). This is one example and not evidence of the average quality.

Conclusion

Avocado prices can be estimated to within about $0.21 on average, but a tuned Random Forest is not clearly better than a simple Linear Regression.

In the Random Forest, the type (organic or conventional) and the sales volumes are the most important features, while calendar and region features are less important. The time-based split gave an honest test: the newest weeks have a higher price level than the training weeks, and all models underestimated the high prices of autumn 2017.

Limitations

- The model uses the sales volumes of the same week as inputs. These numbers are not known in advance, so this is an estimate from market data, not a true forecast of future prices.
- The price level shifts over time and tree models cannot extrapolate outside the values seen in training (for example the year 2018).
- Only one time split was used, so the result depends on this one test period (August 2017 to March 2018), which includes an unusual price spike.
- The data have no information about supply, weather, imports, holidays or costs, and cover retail scanner data from 2015 to March 2018 only.
- Overlapping regions (such as TotalUS and West) mean that rows are not fully independent.
- Only 12 random-search trials and a small parameter grid were used.

Future work

- Add weather, supply and holiday features, and lagged prices from earlier weeks.
- Evaluate on several time-based folds (rolling-origin evaluation).
- Try gradient boosting (XGBoost or LightGBM) and models for each type or region.
- Serve the model through a FastAPI endpoint.

Reproducibility. Code: `project3_avocado_price_predictor.ipynb` (random seed 42). Python libraries: pandas, scikit-learn, matplotlib and seaborn.

Source of the data: Hass Avocado Board weekly retail data, as published on Kaggle ("Avocado Prices").